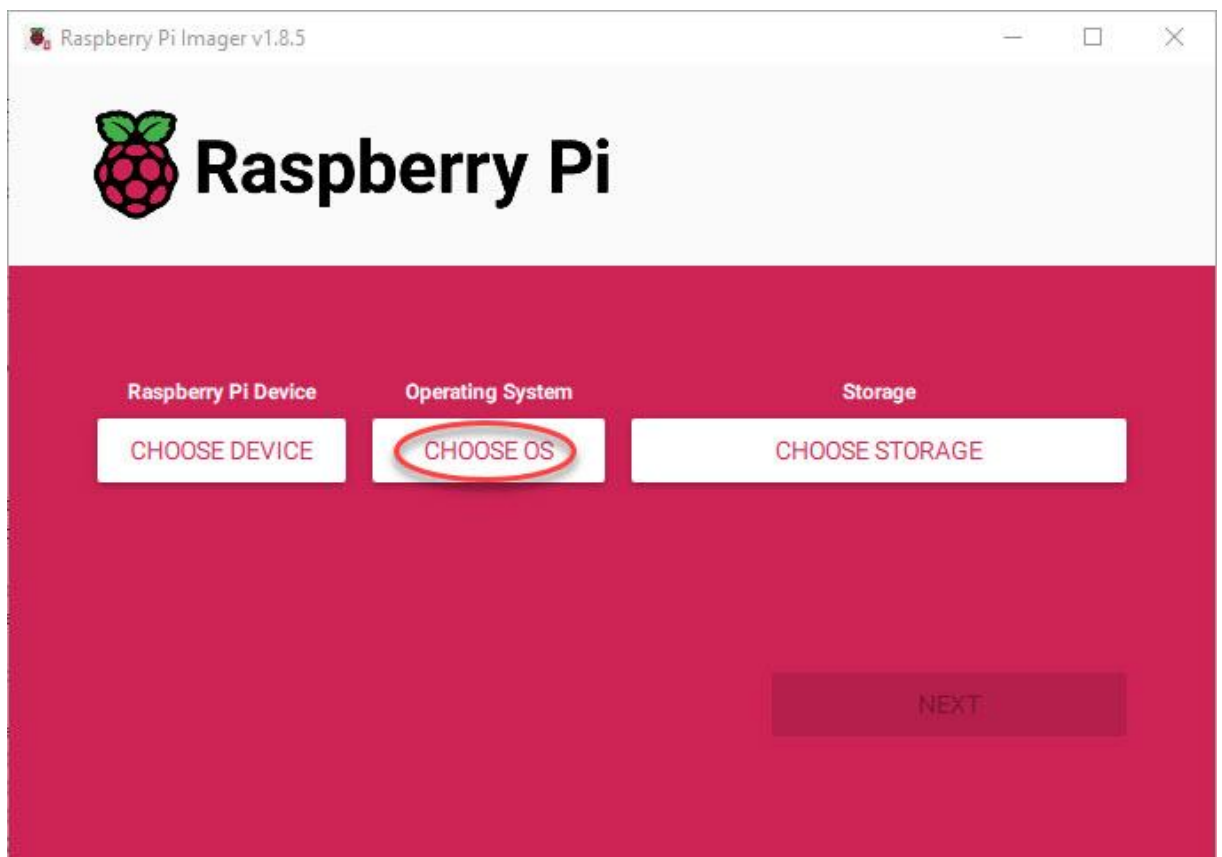
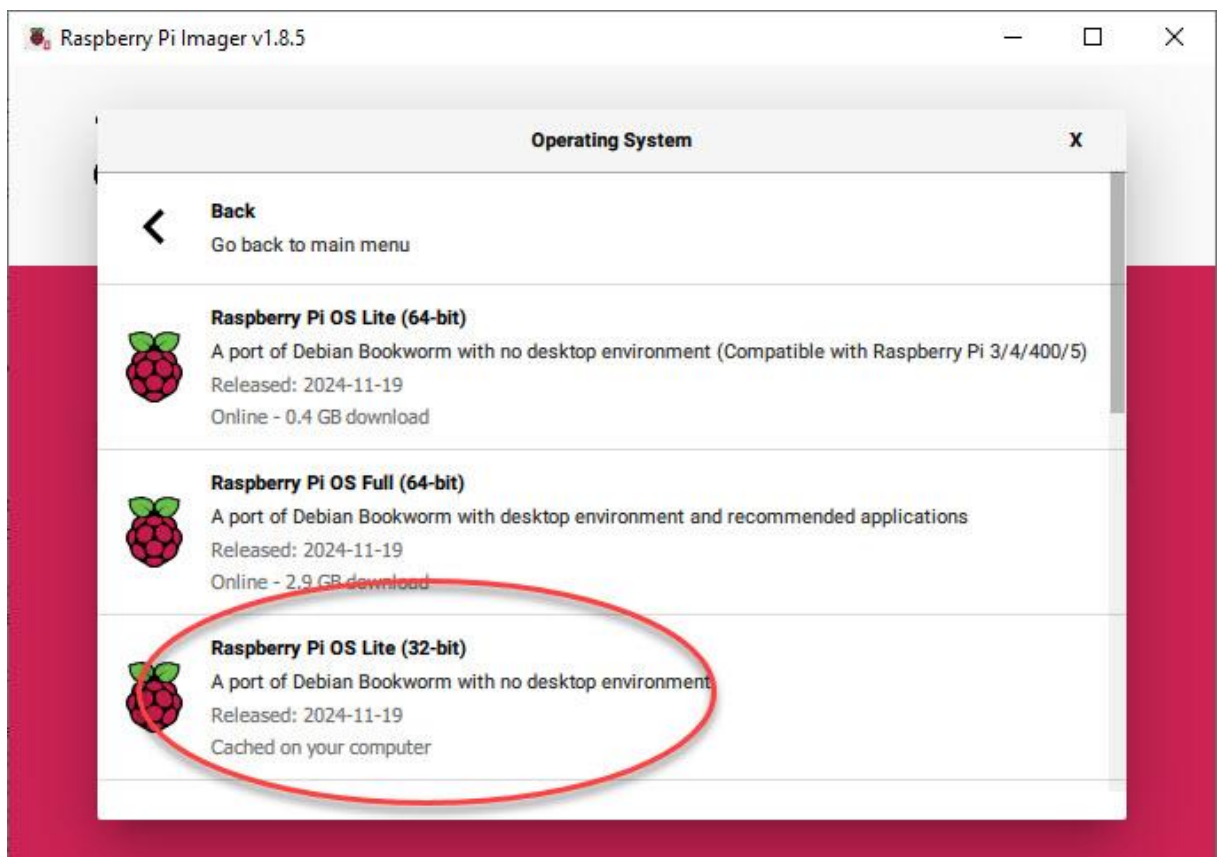
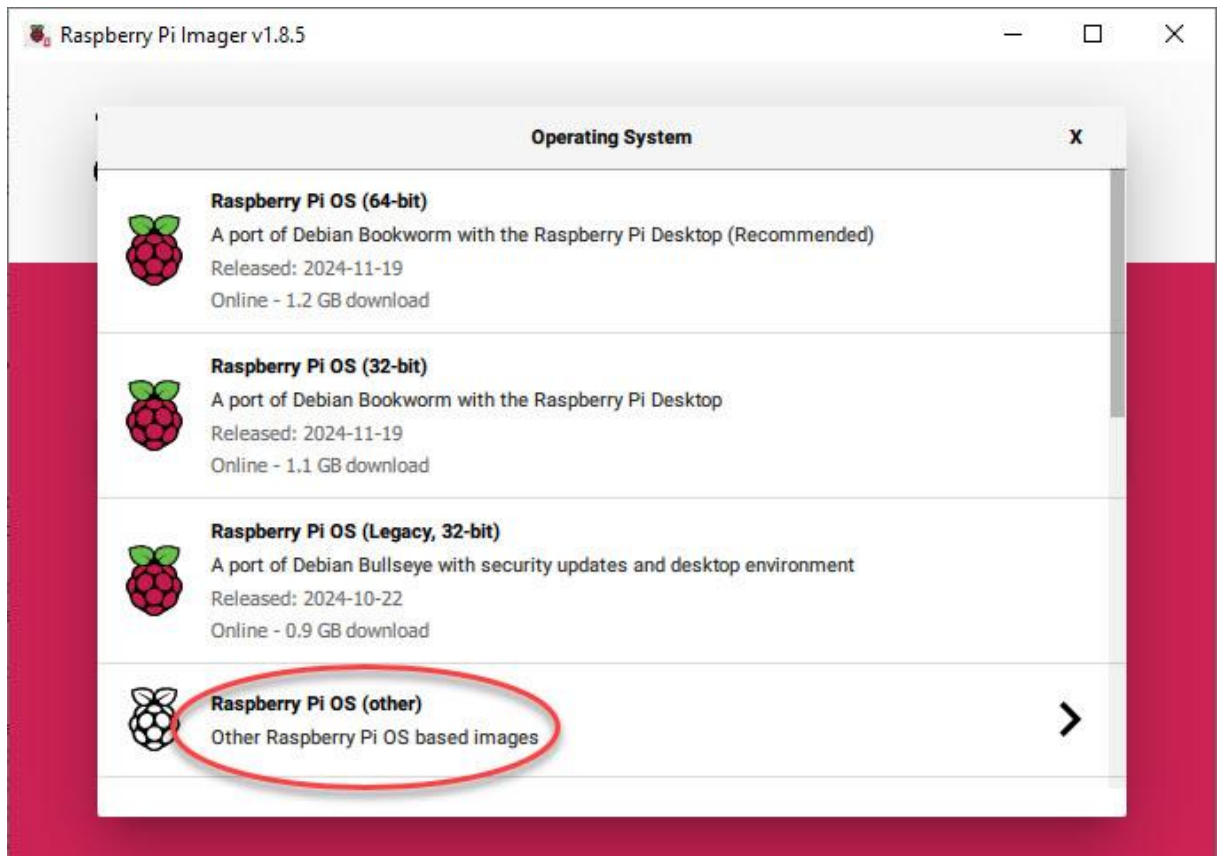


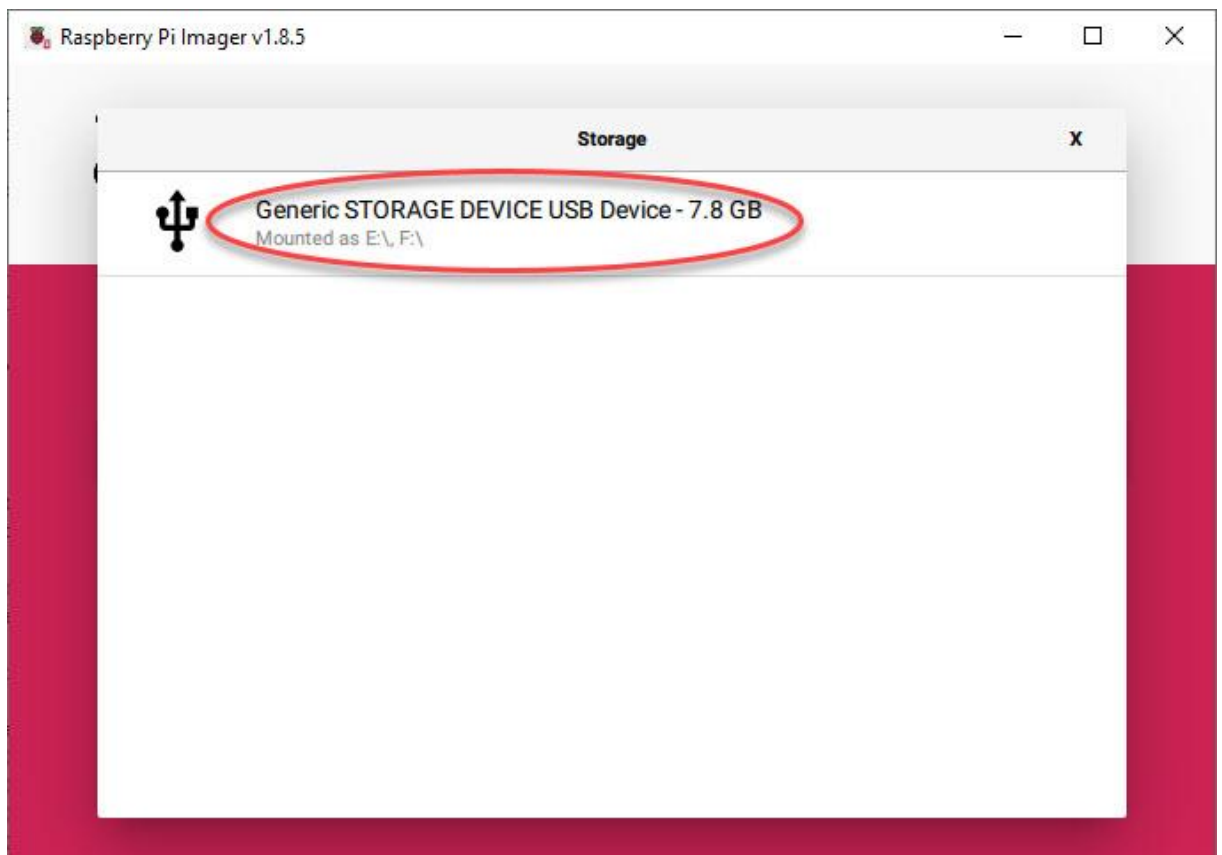
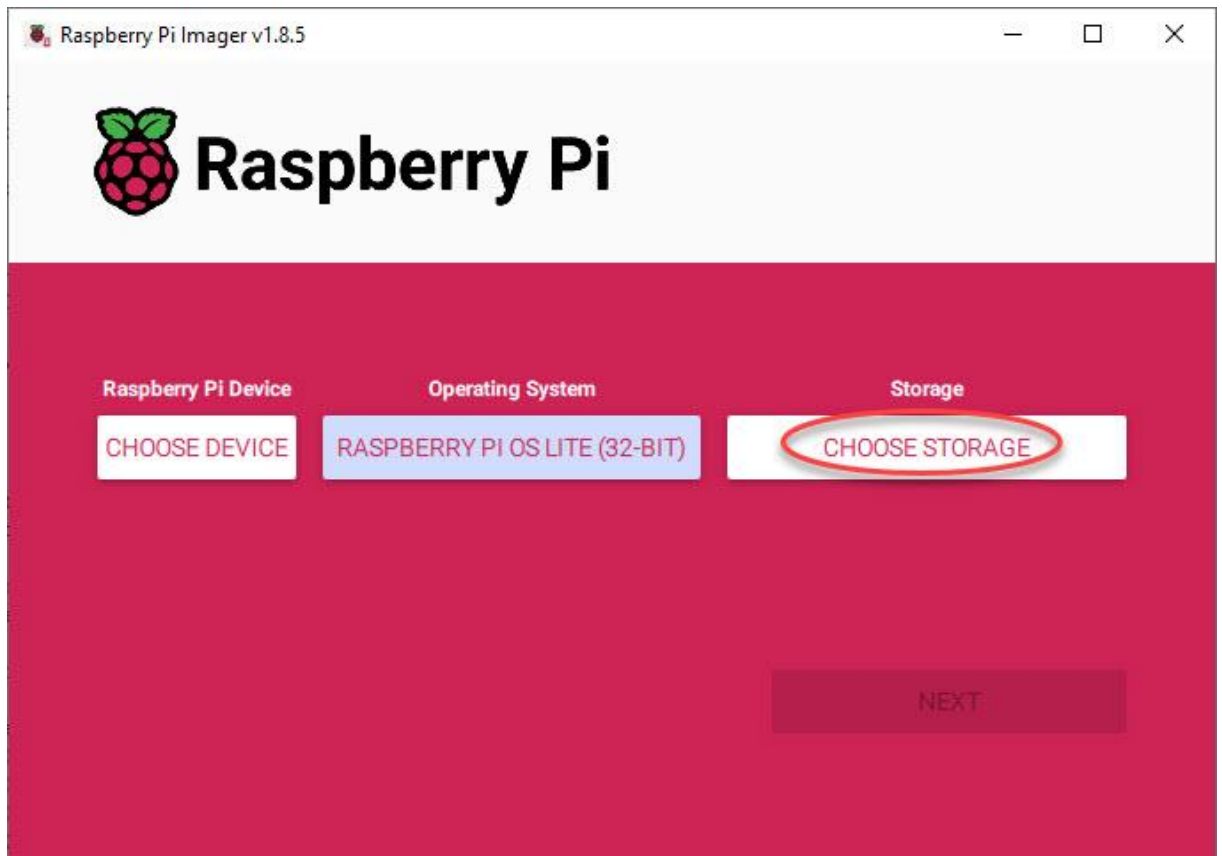
Pi-Ager install on Raspberry Pi 3/Pi 4/Pi 5/Pi zero (2)w under Pi OS 12 Lite bookworm

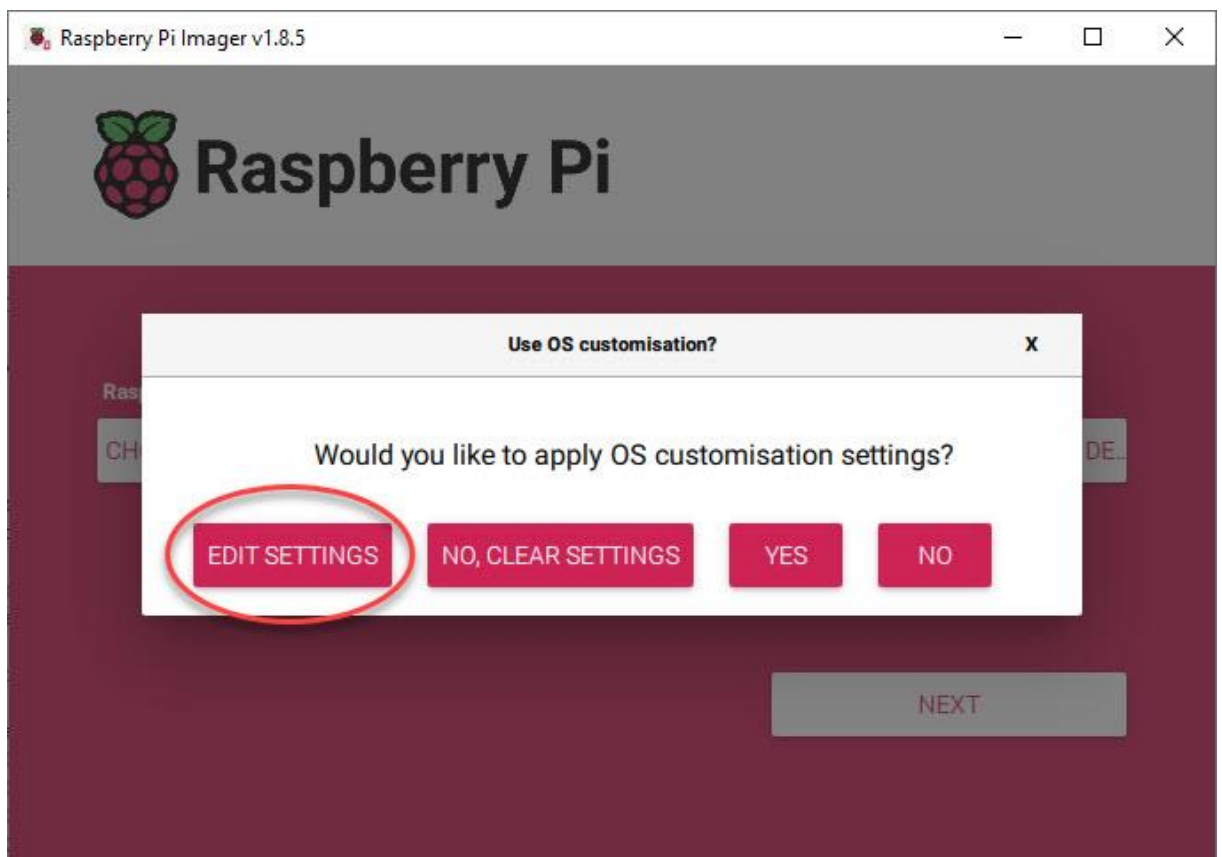
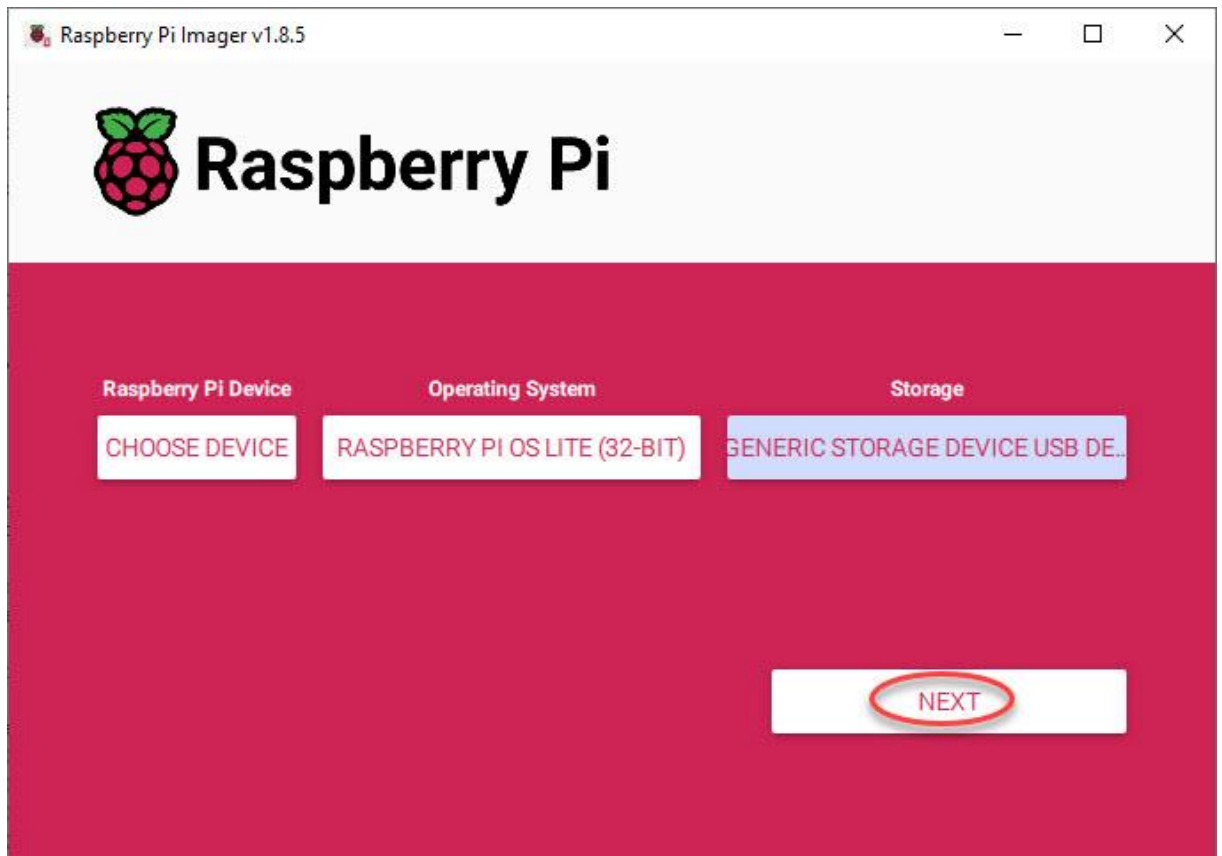
- **For all Pi:**
Download and install Raspberry Pi Imager v1.8.4 or later from <https://www.raspberrypi.com/software/>.

Start Raspberry Pi Imager, then continue with the follow steps:









GENERAL

SERVICES

OPTIONS

☒ Set hostname: .local☒ Set username and passwordUsername: Password: ☒ Configure wireless LANSSID: Password: ☒ Show password ☐ Hidden SSIDWireless LAN country: ▼☒ Set locale settingsTime zone: ▼Keyboard layout: ▼

SAVE

OS Customisation

GENERAL

SERVICES

OPTIONS

☒ Enable SSH

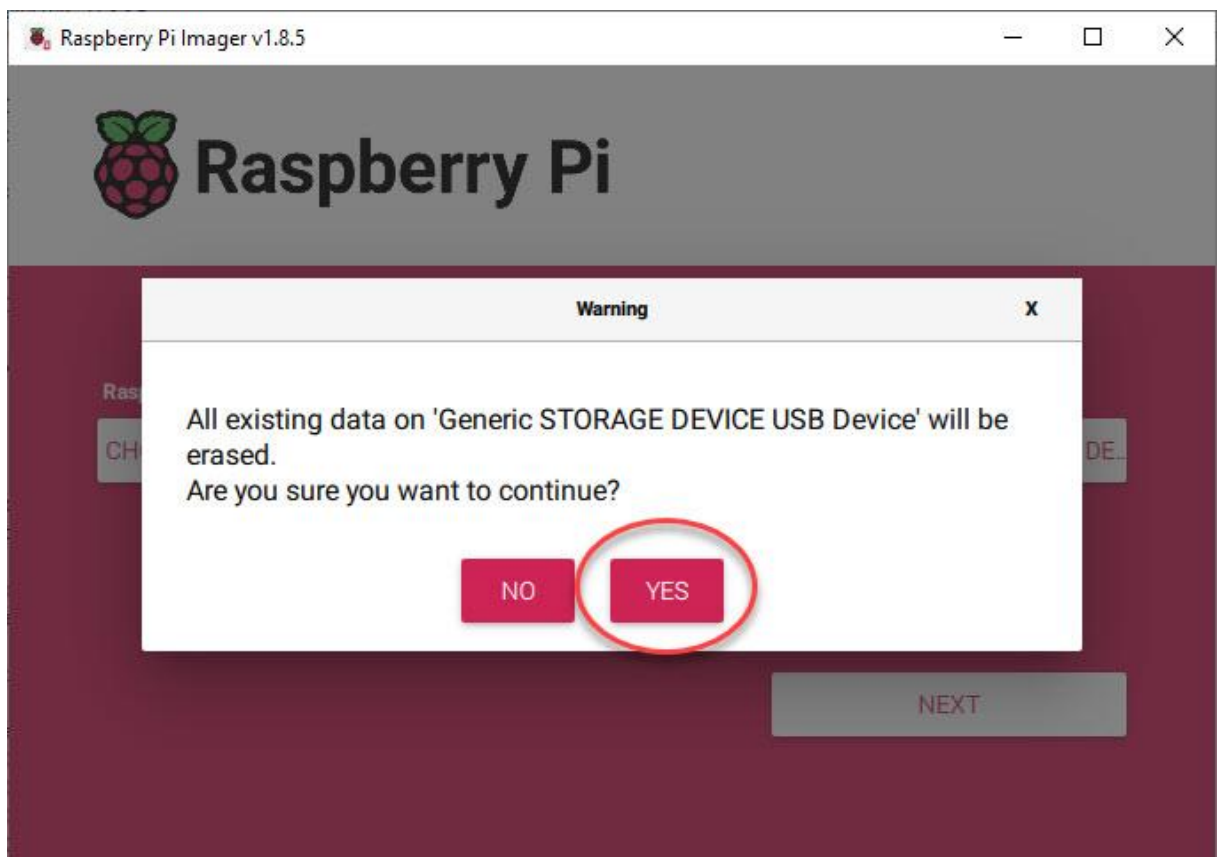
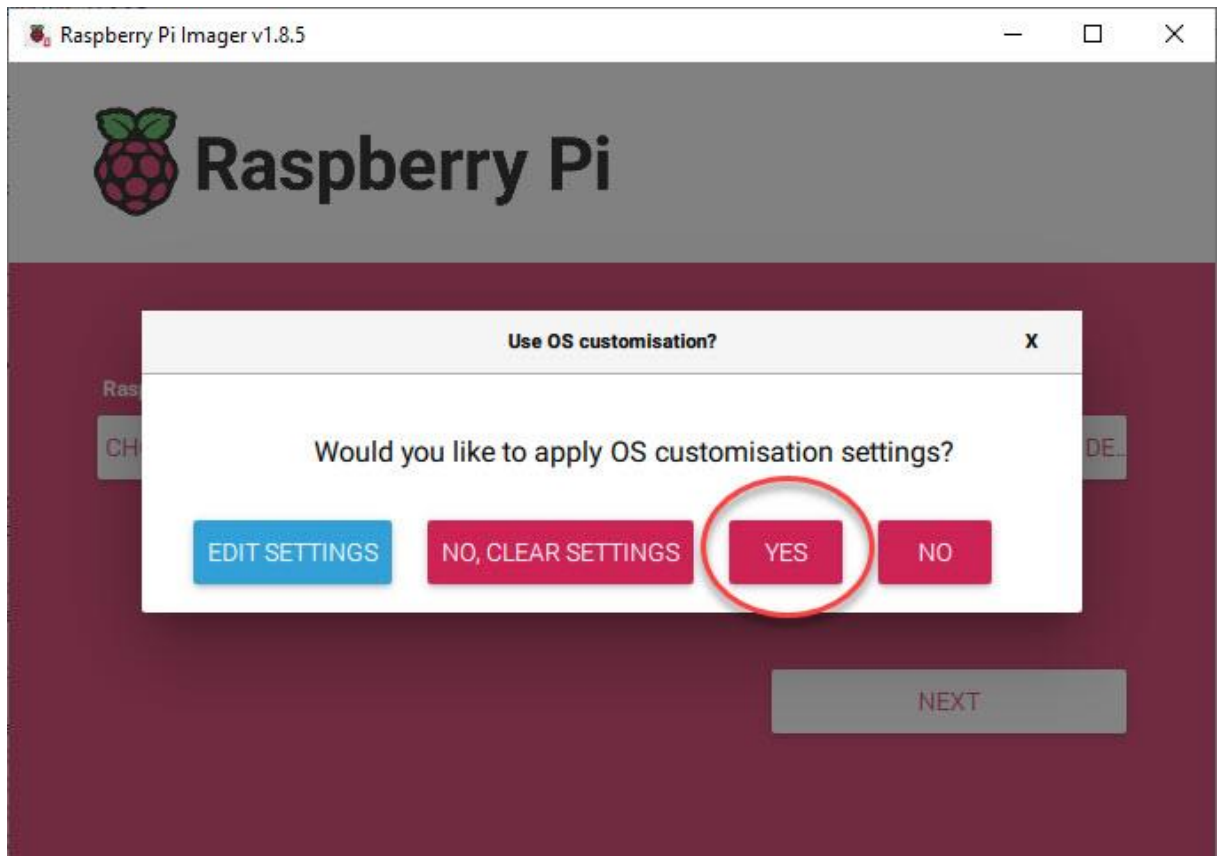
☒ Use password authentication

☐ Allow public-key authentication only

Set authorized_keys for 'pi':

RUN SSH-KEYGEN

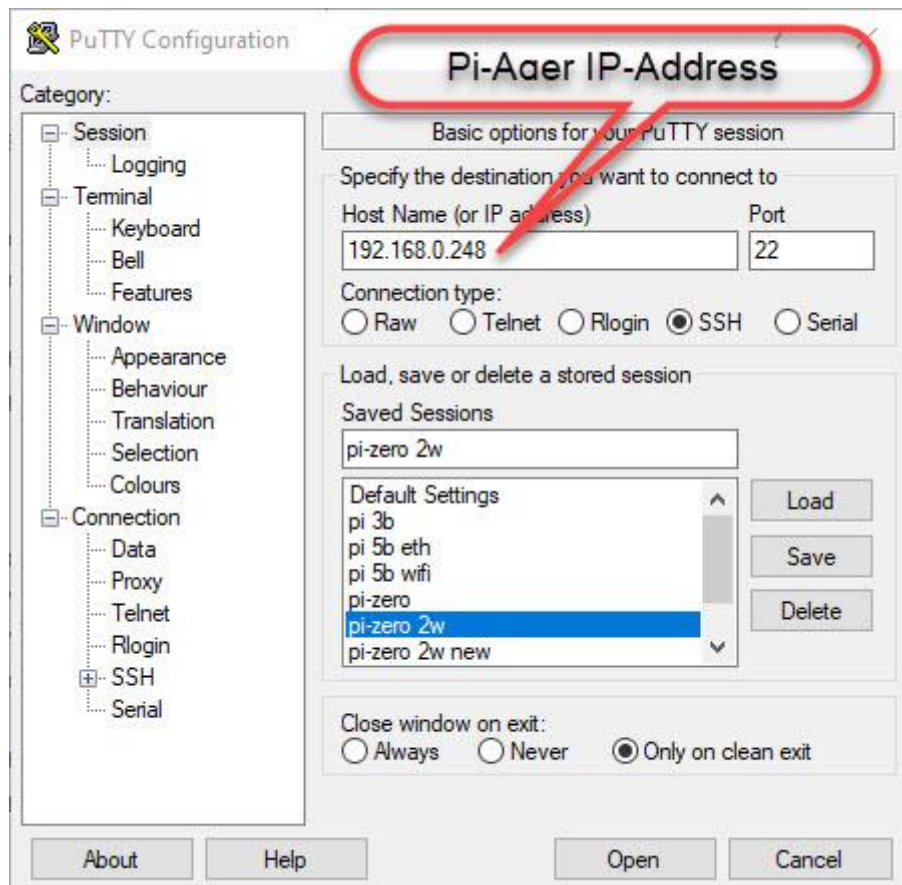
SAVE



Then 'SAVE' and apply OS customization settings with 'YES'.
The customized OS is now written to your SD-Card.

Put your SD-Card in your Raspberry Pi and power on your Pi device.
Login via SSH (e.g. use PuTTY) or connect a HDMI monitor and USB keyboard to your Raspberry device and continue with the setup as described below.

- Download and install PuTTY from <https://www.putty.org/>
- Optional download and install WinSCP from <https://winscp.net/eng/index.php>
- Start PuTTY to connect to Pi-Ager:




```
pi@pi-ager: ~  
login as: pi  
pi@192.168.0.206's password:  
Linux pi-ager 6.6.74+rpt-rpi-v7 #1 SMP Raspbian 1:6.6.74-1+rpt1 (2025-01-27) arm  
v7l  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Wed Mar  5 20:00:41 2025 from 192.168.0.201  
  
SSH is enabled and the default password for the 'pi' user has not been changed.  
This is a security risk - please login as the 'pi' user and type 'passwd' to set  
a new password.  
  
pi@pi-ager:~ $
```

- Download system build scripts from github:
`wget -O pi-ager-build-system.sh -nv https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/pi-ager-build-system.sh`

`wget -O pi-ager-finalize-build.sh -nv https://github.com/Tronje-the-Falconer/Pi-Ager/raw/refs/heads/resources/anleitungen/pi-ager-finalize-build.sh`

`chmod +x *.sh`

Start now first phase of Pi-Ager System setup:

`sudo ./pi-ager-build-system.sh`

You can now follow the system setup, that is divided in 4 steps with rebooting between the steps. Restart the script after rebooting to continue the system setup :

`sudo ./pi-ager-build-system.sh`

After step 3 you should edit the file `/boot/firmware/setup.txt` that contains information about passwords, WLAN setup parameter, temperature/humidity sensor to use and optional HMI display version connected to your Pi-Ager:

`sudo nano /boot/firmware/setup.txt`

```
pi@pi-ager: ~
GNU nano 7.2 /boot/firmware/setup.txt
# ACHTUNG
# Im Router bzw bei den WLAN Security Einstellungen sollte ein Verschlüsselungsverfahren
# WPA2 (CCMP) oder WPA2+WPA3 ausgewählt werden, da sonst keine WLAN Verbindung zwischen
# Pi-Ager und Router bzw. Accesspoint aufgebaut werden kann.
# WPA wird nicht vom Pi-Ager unterstützt
#-----

# WLAN Netzwerkname:
wlanssid=' '

# WLAN Schlüssel (mindesten 8 Zeichen):
wlankey=' '

# ISO code your country, e.g. DE, GB:
country='DE'

# Temperature/Humidity sensor attached to pi-ager:
# Supported types are : SHT75, SHT85, SHT3x, SHT3x-mod, AHT1x, AHT1x-mod, AHT2x, AHT30, SHT4x-A, S
sensor='SHT75'

# hmi TFT display device Namen:
# gültige device Namen: NX3224K028 or NX3224T028 or NX3224F028
# z.B. : hmidisplay='NX3224K028'
# Wenn hmidisplay definiert ist, wird die Firmware des hmi TFT Displays beim ersten Booten geflash
# Das Flashen des Displays dauert bis zu 5 Minuten, der Fortschritt des Flashens kann nur auf dem
# TFT-Display beobachtet werden.
# Während des Flashens sollte die Stromversorgung des Raspi nicht ausgeschaltet werden.
#
hmidisplay=

# setup.txt nach boot löschen? leer lassen für löschen oder 1 für behalten
keepconf=1

```

^G Hilfe ^O Speichern ^W Wo ist ^K Ausschneiden ^T Ausführen ^C Position
^X Beenden ^R Datei öffnen ^\ Ersetzen ^U Einfügen ^J Ausrichten ^/ Zu Zeile

After saving setup.txt the last step initializes the Pi-Ager system with data from setup.txt by starting the script pi-ager-finalize-build.sh:

sudo ./pi-ager-finalize-build.sh

The system is then rebooted and ready to start the Pi-Ager. Open your browser window and enter the IP-Address of the Pi-Ager in the address line.

How to generate and deploy Pi-Ager Images

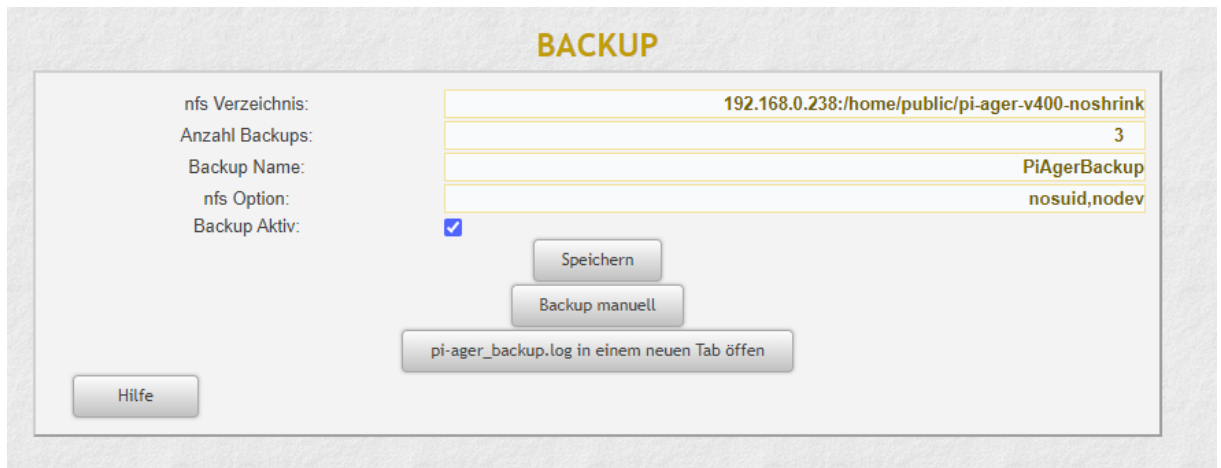
There are some steps to do for generating and deploying a new image from a running Pi-Ager system:

1. Setup a NFS Server to store a new image. The best way is using a Raspi 4B or 5 equipped with an USB Memory stick with 128GB or more. Using RPi 5 it is now possible to add a NVME SSD so that there will be enough space for storing one or more images.

How to setup a NFS Server :

<https://www.elektronik-kompendium.de/sites/raspberry-pi/2007061.htm>

2. On the Pi-Ager setup and start an Image backup on the ADMIN page. For example :



The screenshot shows the 'BACKUP' configuration page in the Pi-Ager ADMIN interface. It features a form with several input fields and a table-like structure for configuration. The fields are labeled on the left: 'nfs Verzeichnis:', 'Anzahl Backups:', 'Backup Name:', 'nfs Option:', and 'Backup Aktiv:'. The corresponding values are entered in the fields on the right: '192.168.0.238:/home/public/pi-ager-v400-noshrink', '3', 'PiAgerBackup', and 'nosuid,nodev'. Below these fields are three buttons: 'Speichern', 'Backup manuell', and a button that says 'pi-ager_backup.log in einem neuen Tab öffnen'. There is also a 'Hilfe' button in the bottom left corner.

Label	Value
nfs Verzeichnis:	192.168.0.238:/home/public/pi-ager-v400-noshrink
Anzahl Backups:	3
Backup Name:	PiAgerBackup
nfs Option:	nosuid,nodev

Buttons: ☒ Backup Aktiv, , , ,

3. Next start a terminal on the Pi-Ager and start the pi-ager_image.sh script with option -l: (lowercase 'L')

```
pi@pi-ager: /usr/local/bin

pi@pi-ager:~$ cd /usr/local/bin/
pi@pi-ager:/usr/local/bin$ sudo pi-ager_image.sh -l
Backup ist inaktiv. Pi-Ager_image wird gestartet!
überprüfe ob der NFS-Server vorhanden ist.
Checking...
192.168.0.238:/home/public/pi-ager-v400-noshrink ist vorhanden
überprüfe ob Backup Pfad vorhanden ist.
Checking ...
/home/nfs/public ist vorhanden
überprüfe ob NFS Mount point definiert ist.
Checking ...
/home/nfs/public ist definiert
überprüfe ob NFS Mount point im file system angelegt ist.
Checking...
/home/nfs/public ist angelegt
NFSVOL=192.168.0.238:/home/public/pi-ager-v400-noshrink
NFSPATH=/home/nfs/public
NFSMOUNT=/home/nfs/public
umount: /home/nfs/public: not mounted.
hänge NFS-Volume 192.168.0.238:/home/public/pi-ager-v400-noshrink ein
mount with options: nosuid,nodev
mount NFS-Volume 192.168.0.238:/home/public/pi-ager-v400-noshrink successfull. Image script continues
.
Load last backup file.
Backup path is /home/nfs/public
Backup file is /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
Backup path with file is /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
Source File = /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
do_copy = false
my_image = false
Using /home/nfs/public/PiAgerBackup_2024-02-12-103529.img as source and target
#####
parted_output = 2:541065216B:15931539455B:15390474240B:ext4::;
partnum = 2
partstart = 541065216
loopback = /dev/loop0
#####
parted_output_boot = 1:4194304B:541065215B:536870912B:fat32::lba;
partnum_boot = 1
partstart_boot = 4194304
loopback_boot = /dev/loop1
#####
mount directory is /tmp/tmp.SfM3akt9CW
2024-02-12 10:39:10 URL:https://raw.githubusercontent.com/Tronje-the-Falconer/Pi-Ager/entwicklung/boo
t/firmware/setup.txt [3281/3281] -> "setup.txt" [1]
setup.txt copied to /tmp/tmp.SfM3akt9CW/boot/
cmdline.txt modified, added init=/usr/lib/raspberrypi-sys-mods/firstboot
rematch1 /tmp/
rematch2 tmp.SfM3akt9CW
change rootdir is tmp.SfM3akt9CW
/tmp
pi-ager_main.service disabled, will be enabled during setup_pi-ager.sh
Created symlink /etc/systemd/system/multi-user.target.wants/setup_pi-ager.service -> /lib/systemd/system/setup_p
i-ager.service.
setup_pi-ager service enabled
Running in chroot, ignoring command 'is-active'
hostname changed to pi-ager
unmount /tmp/tmp.SfM3akt9CW/boot
unmount /tmp/tmp.SfM3akt9CW
Detaching loop devices from /home/nfs/public/PiAgerBackup_2024-02-12-103529.img
The image /home/nfs/public/PiAger_image_2024-02-12-103940.img was successfully created.
pi@pi-ager:/usr/local/bin$
```

4. Zip and deploy your new image.